

Capstone Design Manager

Iteration 1 Review



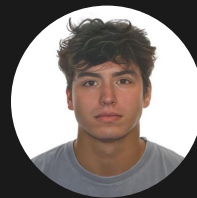
Gaurav Shrivastava



Jayden White



Margulan Baizhakyp



Noland Miller

Sponsor Dr. Monica Anderson Herzog, Computer Science, The University of Alabama



Margulan

Iteration 1 Goal and sponsor input

Iteration Goal

- The inherited system running on every teammate's machine and in CI
- Architecture decisions recorded, not just discussed
- First sponsor-facing feature demonstrable: project submission

What the sponsor told us (email, 9/22)

- Treat assignment as a constraint-satisfaction problem: the computer proposes, the admin approves
- Suggested Google OR-Tools CP-SAT (Python), cloud solver also acceptable
- "The problem size is fairly small, even for 100 students and 25 projects"

2.9×10^{123}

possible ways to put 100 students onto 25 projects of 4. Too many to check one by one, so a solver searches smartly.

She did not weigh in on the other questions (ranking rule, teammate preferences). We will confirm those before Iteration 2 planning.

Decisions made this iteration



Margulan

ADR-001 Build on v1

- Keep Vue 3 + Django REST + PostgreSQL and the prior team's code
- Changed from the preliminary "ground up" plan
- Why: it runs, 188 tests pass, and the missing features are new work either way
- Cost: a cleanup pass (dead code, unpinned deps)

ADR-002 Auth

- Keep Auth0 for login, on a tenant we control
- Roles decided server-side: student = on roster, sponsor = everyone else, admin = granted by admin
- Add role checks to the API; authenticate the two open endpoint groups
- MySSO deferred to handoff recommendations

Data model & dev environment

- ERD of the current schema plus a table of planned changes (role, capacity 4, constraints, server-side deadline)
- Docker Compose runs the full stack; guide in docs/development-guide/
- Noland built the environment and CI fix; out sick today

Decisions, written down



Margulan

PreviewCodeBlame28 Lines (18 loc) · 2.55 KB

RawDownloadEditMenu

ADR-001: Build on the v1 codebase rather than rebuild

Status: Accepted Date: 2026-09-22 Deciders: Gaurav Shrivastava, Jayden White, Margulan Baizhakyp, Noland Miller

Context

The sponsor handed us the Spring 2026 team's working application: Vue 3 + Vite frontend, Django REST Framework backend, PostgreSQL, Auth0 login, MiniO/S3 storage, Docker Compose. It runs, has 188 passing backend tests, and implements the core semester workflow (sponsor submission, student ranking, manual assignment, sponsor emails). Its gaps are documented in [docs/investigations/v1-reference.md](#).

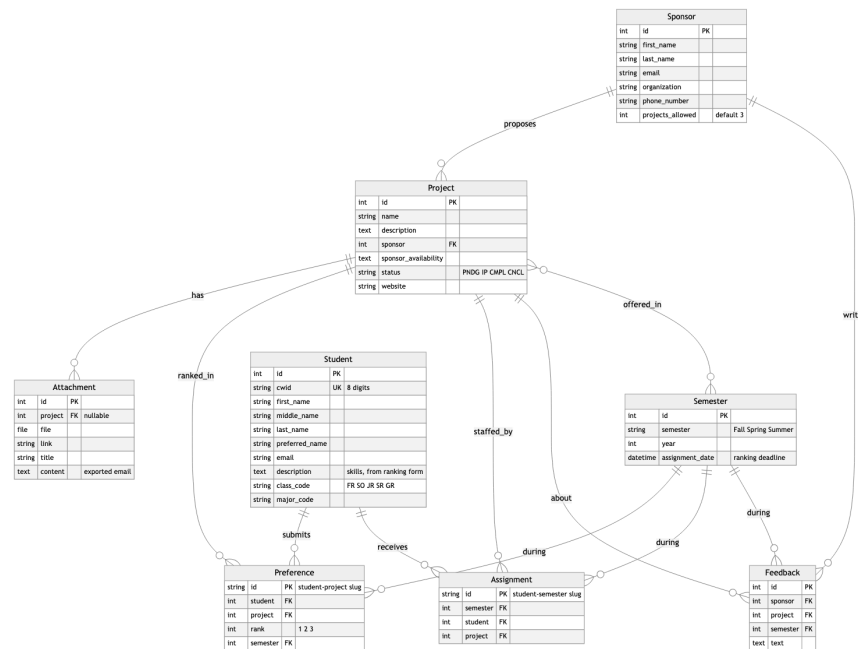
The course requires a live demo of integrated working software at each iteration, the first on 9/24, and a full handoff at semester end. Four developers, roughly ten working weeks.

Options considered

- Build on v1. Keep the stack and the code. Fix the security gaps, delete dead code, pin dependencies, add the features the sponsor needs. Tradeoff: we inherit some design choices we would not make ourselves, and the codebase has rough edges (unpinned dependencies, dead components, hardcoded values).
- Rebuild on the same stack. Same technologies, fresh code, using v1 only as a reference. Tradeoff: the first two iterations reproduce

docs/decisions/ADR-001-stack.md on GitHub

Both on main, both linked from the README. A future team reads why, not just what.



docs/development-guide/data-model.md: current schema (ERD)

Plan vs. actual



Jayden

11 / 17

Iteration 1 Issues closed

6

carried forward, all visible in the
Results view

*Nothing removed from the plan;
incomplete items stay In Progress with
the reason in the Issue.*

Capstone Design Manager

View 1

01 - Product Backlog

02 - Iteration 1 Plan

03 - Iteration 2 Plan

04 - Iteration 3 Plan

05 - Iteration 1 Results

06 - Iteration 2 Results

07 - Iteration 3 Results

Iteration:"Iteration 1"

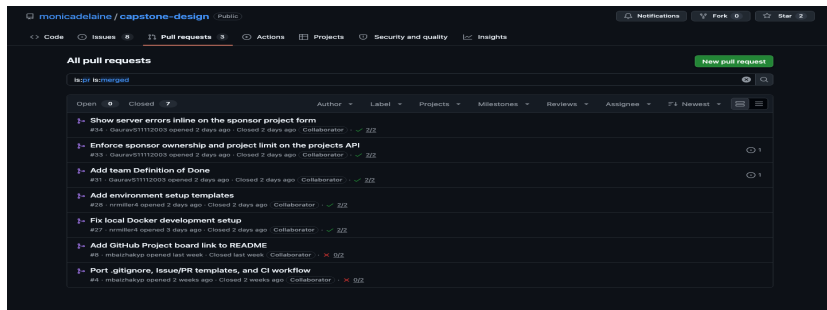
17

View

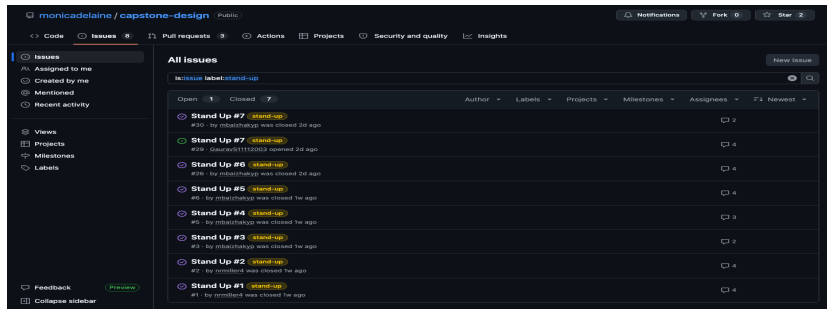
Title	Assignees	Status	Linked pull requests	Sub-issues progress
1 SPIKE-01: Decide the technology stack (keep v1 or rebuild) #9	nrmiller4	Done		
2 SPIKE-02: Decide the authentication approach #10	nrmiller4	Done		
3 DOC-03: Design the data model for the new system #11	nrmiller4	Done		
4 SETUP-04: Development environment and development guide #12	nrmiller4	In Progress		
5 SETUP-02: Configure the GitHub Project board #13	jtwhite18	Done		
6 PLAN-01: Create and prioritize the Product Backlog #14	jtwhite18	Done		
7 PLAN-03: Capture the Iteration 1 planning snapshot #15	jtwhite18	Done		
8 PBI-02: Student ranks projects (preference form) #16	jtwhite18	In Progress		
9 PBI-01: Sponsor submits a project #17	GauravS11112003	Done	#33	
10 TEST-01: Establish the automated test baseline #18	GauravS11112003	In Progress	#35	
11 SETUP-05: Make CI pass on every pull request #19	GauravS11112003	In Progress	#36	
12 PLAN-02: Define the team Definition of Done #20	GauravS11112003	Done	#31	
13 SETUP-03: Rewrite the README as the project landing page #21	mbaizhakyp	Done		
14 DOC-01: Establish the docs/ handoff structure #22	mbaizhakyp	Done		
15 DOC-02: Establish the course/ process record structure #23	mbaizhakyp	In Progress		
16 PLAN-04: Sponsor planning record and Iteration 1 Goal #24	mbaizhakyp	Done		
17 DOC-04: Iteration 1 presentation, review, retrospective, and tag #25	mbaizhakyp	In Progress		

GitHub Project: 05 - Iteration 1 Results view, captured 9/24

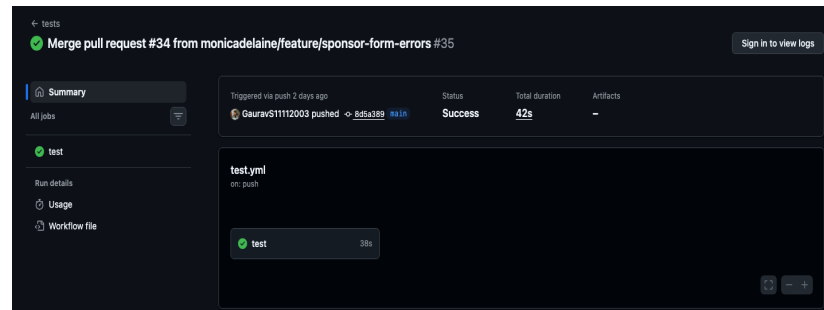
GitHub evidence



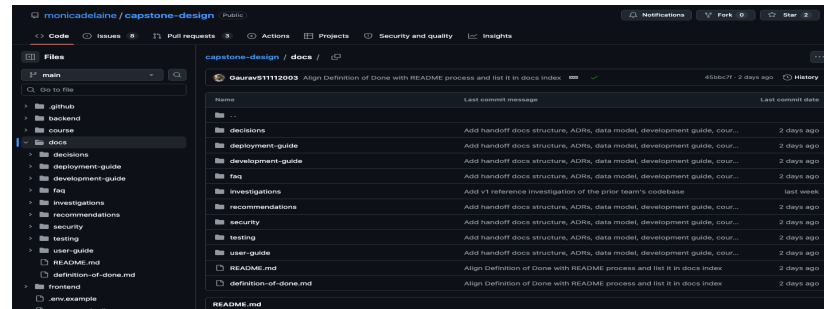
Pull requests: #4, #8, #27, #28, #31, #33, #34 merged with review



7 stand-up Issues, one comment per person per meeting



GitHub Actions: tests on every push and PR, green on main



docs/: decisions, development guide, investigations, placeholders

Backlog changes and Iteration 2

JW

Jayden

Added because of what we learned

- SPIKE: evaluate OR-Tools CP-SAT against our data model (sponsor request)
- PBI: solver-proposed assignment, admin reviews and approves
- SETUP: our own Auth0 tenant; remove the client-side role hint
- SECURITY: role permission class; authenticate semester and email endpoints
- DEPLOY: hosting investigation; prior team's cloud setup is gone

Iteration 2 focus (proposed)

- Student ranking form finished and tested (carried)
- Security fixes from ADR-002
- OR-Tools spike result, then the assignment feature
- A deployed instance the sponsor can open
- Confirm ranking rule and teammate preferences with the sponsor first

Retrospective



Margulan

Keep

- PR + review for code, direct commits for docs: fast and still reviewed
- Stand-up comments in the required format; evidence links made progress visible

Change

- Blockers owned by someone else get escalated at the next stand-up, not carried
- One person owns the board; everyone else adds items
- Every setup change is verified on both platforms before the Issue closes

Problem

- Three commitments carried for three weeks waiting on one PDF export
- Two boards with the same name; a week to sort out access
- Mac/Linux setup broke on a file permission nobody on Windows saw

Action (owner)

- Preliminary PDF committed by Friday (Jayden)
- Each teammate posts "runs on mine" on #12 by Friday (all)
- Sponsor answers on ranking rule and teammates before Iteration 2 planning (Margulan)

Links

Repository	github.com/monicadelaine/capstone-design
Project board	github.com/users/mbaizhakyp/projects/1
Decisions	docs/decisions/ (ADR-001, ADR-002)
Iteration 1 records	course/iterations/iteration-1/
Stand-ups	Issues labeled stand-up
Tag	iteration-1-submission

Quality



Gaurav

188

backend tests passing

every PR

runs the suite in GitHub Actions

DoD

defined and linked from the README

The screenshot shows the GitHub Actions interface for a workflow named 'test'. The top bar indicates the workflow was triggered by a push to the 'main' branch. The status is 'Success' and the total duration is '42s'. The 'test' job is shown as completed with a duration of '38s'.

Latest run on main: success, 42 s

Definition of Done

- Acceptance criteria met
- Tests pass
- Docs updated
- PR reviewed, merged
- AI use disclosed
- Demonstrable

Live demo



Gaurav

1

Sponsor logs in and submits a project

form → API → database row

2

Project appears in the project list

the sponsor-facing slice from PBI-01

3

Django admin shows the project and the semester

where the instructor manages the term today

4

Test run or CI badge

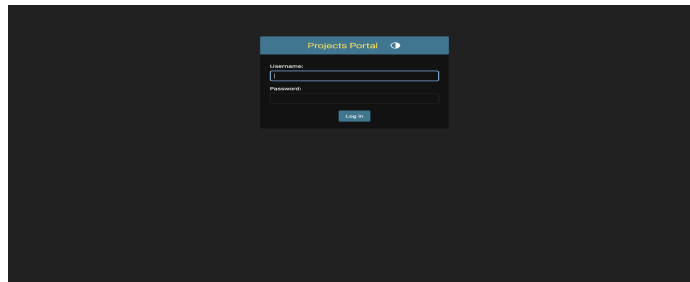
188 passing, green on main

Capstone Project Manager

Login

Sign Up

localhost:5173, the app running locally



Django admin, where the semester is managed